

Executive Summary

This study compares two optimization strategies for maximizing the lift-to-drag ratio of NACA 4-digit airfoils: **Reward-Directed Diffusion (RDD)** and **Bayesian Optimization (BO)**. Both methods use the same 50 Sobol-distributed starting airfoils, common parameter bounds, and the same trained artificial neural network surrogate for aerodynamic guidance.

Surrogate validation	Held-out test $R^2 = 0.9984$ for CL and 0.9924 for CD.
Best RDD result	True XFOIL L/D = 184.27 at m7.5, p4.5, t10.3.
Best BO result	ANN-predicted L/D = 183.76 at m7.5, p4.7, t10.3; reached at evaluation 67.
Shared design trend	Both methods converged to maximum camber, camber near mid-chord, and thickness near 10–11%.
Main caution	A strict winner cannot be declared until the best BO design is verified with the same true XFOIL procedure used for RDD.

1. Common Experimental Framework

- Design variables: maximum camber m , camber location p , and thickness t .
- Physical bounds: $m = 0.015\text{--}0.075$, $p = 0.15\text{--}0.75$, and $t = 0.05\text{--}0.20$.
- Aerodynamic condition: Reynolds number 1.5×10^6 and angle of attack 5° .
- Objective: maximize $L/D = CL/CD$.
- Starting set: the same fixed CSV containing 50 Sobol-distributed airfoils.
- Surrogate dataset: 2,000 Sobol-sampled airfoils evaluated with XFOIL.
- Surrogate architecture: 800 coordinate inputs, shared hidden trunk, and separate CL and $\log(CD)$ output heads.
- Designs are rounded on the NACA digit scale before scoring and saving.

2. Surrogate ANN Validation

The ANN was evaluated on an independent held-out test set. **The resulting parity plots show excellent agreement with XFOIL: $R^2 = 0.9984$ for CL and $R^2 = 0.9924$ for CD.** The lift predictions lie almost exactly on the ideal diagonal, while drag predictions also remain tightly clustered around the reference line. These results support using the ANN as a fast reward model during both RDD and BO.

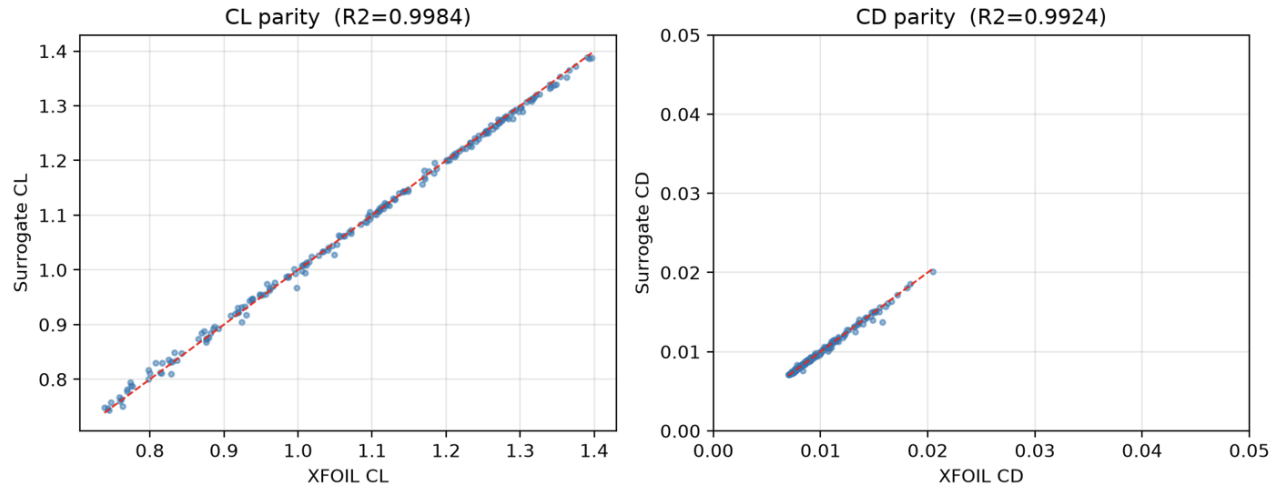


Figure 1. Held-out surrogate parity plots for CL and CD. The dashed line represents perfect agreement with XFOIL.

3. Reward-Directed Diffusion (RDD)

3.1 Method

The generic DDPM was pretrained on 1,000 Sobol-distributed (m, p, t) designs over the complete permitted space. This stage teaches the model a reusable prior over valid design parameters; it does not use aerodynamic rewards. During optimization, a copy of the pretrained model is fine-tuned using surrogate-predicted L/D as a reward. High-reward designs receive larger weights in the DDPM noise-prediction loss. SVDD then guides reverse diffusion by assigning higher sampling probability to candidate denoising steps whose estimated final designs have higher surrogate L/D.

Each of the three rounds sampled 200 trajectories. Generated designs were rounded, deduplicated within the evaluation step, assessed with true XFOIL, and the best verified designs were added back to the fine-tuning dataset for the next round.

3.2 Fine-Tuning Behaviour

The mean surrogate reward increased clearly between rounds: approximately 115 in Round 1, 132–140 in Round 2, and 145–146 in Round 3. The within-round curves fluctuate because batches, diffusion noise, and reward weights are stochastic. The more important pattern is the upward shift between rounds, which indicates that feedback from the best generated designs progressively enriched the fine-tuning dataset.



Figure 2. Average surrogate L/D during DDPM reward-directed fine-tuning across three rounds.

3.3 RDD Results

Round	Saved valid results	Best L/D	Mean L/D	Std. dev.	Minimum L/D
1	109	184.27	179.36	14.44	101.43
2	108	184.18	182.48	1.25	177.05
3	106	184.27	182.56	1.32	175.07

Across all rounds, the output file contains **323 XFOIL-verified rows** and **197 distinct rounded parameter combinations**. The reduction from 600 sampled trajectories reflects repeated rounded designs and the filtering of non-unique or unsuccessful cases. Round 1 contains a few weak designs, producing a large standard deviation. By Rounds 2 and 3, the outputs are tightly concentrated near $L/D \approx 182$ – 184 .

Best RDD design: $m = 0.075$, $p = 0.45$, $t = 0.103$ (digit scale: m7.5, p4.5, t10.3). XFOIL returned $CL = 1.4465$, $CD = 0.00785$, and $L/D = 184.27$

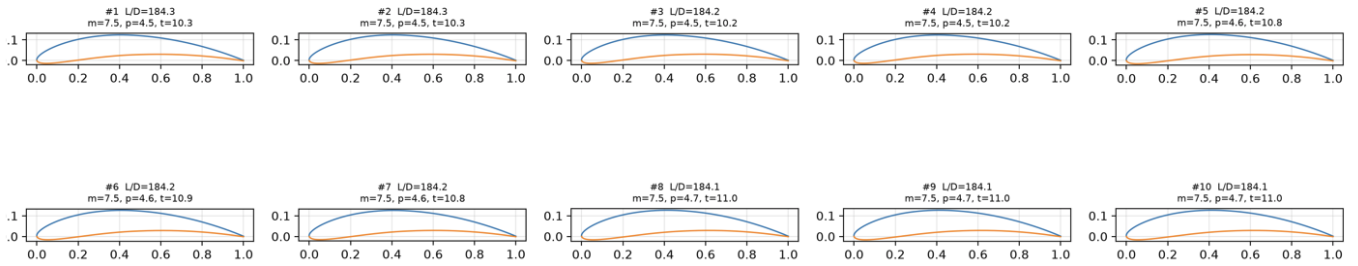


Figure 3. Highest-ranked RDD results across the three rounds. Repeated shapes indicate that the same rounded optimum was rediscovered in multiple rounds.

Interpretation. The leading designs cluster at the maximum permitted camber (m digit 7.5), camber position around 4.5–4.7, and thickness around 10.2–11.0. The narrow cluster indicates convergence toward a compact high-performance region rather than several unrelated optima.

4. Bayesian Optimization

4.1 Method

BO used the same 50-airfoil CSV as a warm start. The ANN supplied CL, CD, and L/D for these initial points. A Gaussian-process model with expected improvement then selected 550 additional continuous designs, giving 600 logged surrogate objective evaluations. Each suggestion was rounded before scoring, so distinct continuous proposals could map to the same rounded design.

4.2 Convergence

The best warm-start airfoil had surrogate L/D = 161.92. The final best BO design was reached at evaluation 67, only 17 acquisition evaluations after the 50-point warm start. The best-so-far curve then remained essentially flat, indicating rapid convergence to the predicted high-performance region. The lower individual points later in the run reflect continued exploration and repeated sampling across the wider bounds.

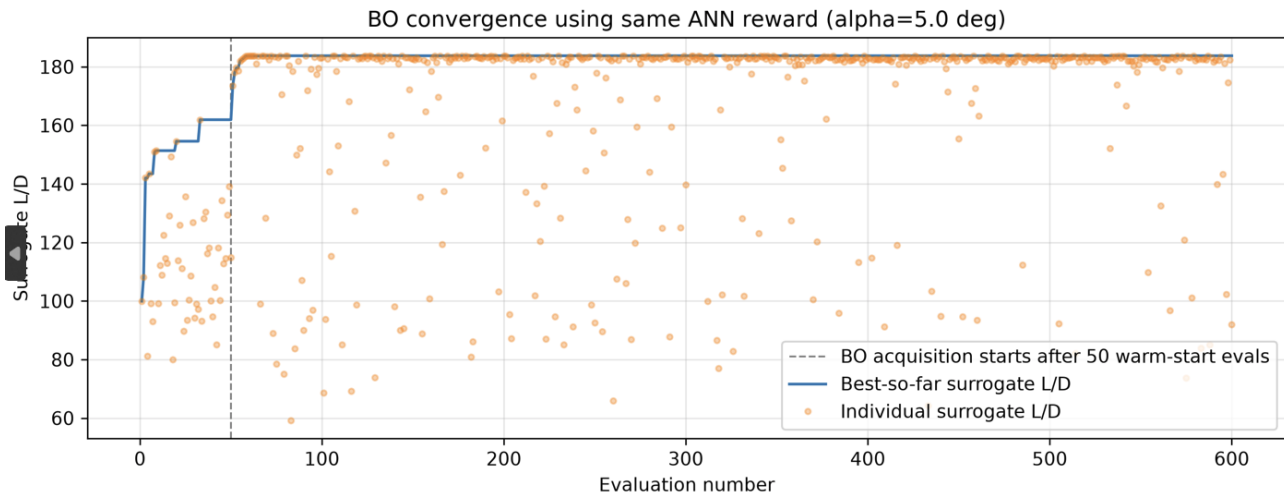


Figure 4. BO convergence using the common ANN reward. The dashed line marks the transition from the 50 warm-start evaluations to GP-guided acquisition.

4.3 BO Result

Metric	BO result
Warm-start observations	50
GP acquisition evaluations	550
Total logged objective evaluations	600
Distinct rounded designs	355
Repeated rounded evaluations	245
Best reached at evaluation	67
Best design (digit scale)	m7.5, p4.7, t10.3
Best ANN L/D	183.76

Best BO design: $m = 0.075$, $p = 0.47$, $t = 0.103$ (digit scale: m7.5, p4.7, t10.3). The surrogate predicted $CL = 1.4495$, $CD = 0.007888$, and $L/D = 183.76$.

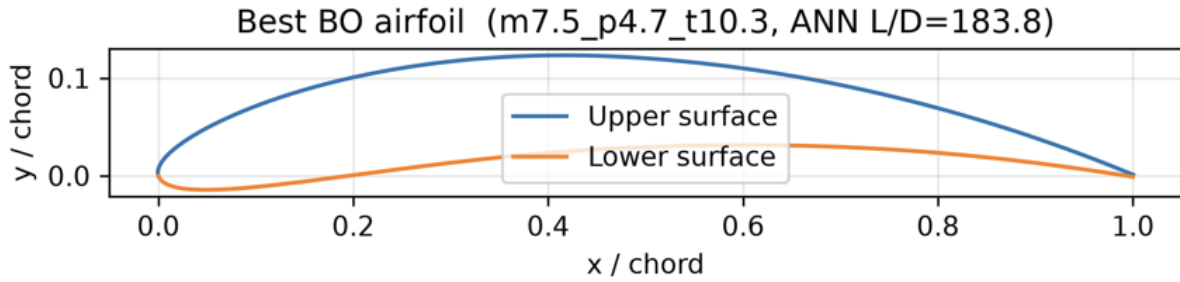


Figure 5. Best BO airfoil according to the ANN surrogate: m7.5, p4.7, t10.3.

5. Comparison of RDD and BO

Aspect	RDD/DDPM	Bayesian Optimization
Initial set	Same 50-airfoil CSV	Same 50-airfoil CSV
Search mechanism	Reward-weighted DDPM fine-tuning + SVDD sampling	Gaussian-process model + expected improvement
Configured search activity	3 rounds $\times$ 200 trajectories	50 warm start + 550 GP acquisitions
Saved/evaluated outputs	323 XFOIL-verified rows; 197 distinct rounded designs	600 ANN evaluations; 355 distinct rounded designs
Best design	m7.5, p4.5, t10.3	m7.5, p4.7, t10.3
Best reported L/D	184.27 (true XFOIL)	183.76 (ANN surrogate)
Convergence pattern	Progressive improvement across rounds	Rapid improvement; final best at evaluation 67

5.1 Main Interpretation

Both methods independently converged to almost the same design region. The maximum camber reached the upper bound in both solutions, thickness converged to 10.3%, and the only meaningful difference was camber location (4.5 for RDD versus 4.7 for BO). This agreement strengthens confidence that the high-performance region is not an artefact of one optimizer alone.

RDD showed round-by-round enrichment: the mean XFOIL L/D increased and the output spread narrowed sharply after Round 1. BO, in contrast, found its final best surrogate design very early after acquisition began and then spent the remaining budget mainly exploring or revisiting rounded regions.

7. Conclusion

The shared surrogate was highly accurate on held-out data, and both optimizers consistently identified a narrow high-performance region near $m7.5$, $p4.5-4.7$, and $t10.3$. RDD achieved a best XFOIL-verified L/D of 184.27 and displayed clear improvement across its iterative rounds. BO converged rapidly to an ANN-predicted L/D of 183.76 after only 17 acquisition evaluations. The results demonstrate that both reward-directed generative optimization and Gaussian-process optimization can effectively exploit the same aerodynamic surrogate.

Results

Quantity	Value
Surrogate test R^2 (CL)	0.9984
Surrogate test R^2 (CD)	0.9924
RDD saved rows	323
RDD distinct rounded designs	197
RDD overall mean L/D	181.45
RDD best true L/D	184.27
BO logged evaluations	600
BO distinct rounded designs	355
BO warm-start best ANN L/D	161.92
BO best ANN L/D	183.76